PhantomSDR-Plus "ORSAT"

User Manual – Web Interface

http://orsat.ddns.net:8080

Complete guide to the online radio-reception interface: bands, modes, filters, audio processing, digital decoders, recording, bookmarks and related tools.

English Version — 2026 Edition

This customized version of PhantomSDR is the result of collaborative work by SV1BTL and F1NSK

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1. General Introduction

PhantomSDR-Plus is a software-defined radio (SDR) receiver accessible from an ordinary web browser, with no installation required. The interface displays the receiver's radio-frequency spectrum and waterfall in real time, and lets you listen to, tune and decode received signals directly from the browser, on both computer and mobile phone, with certain restrictions on the latter.

PhantomSDR aims to be a modern alternative to PA3FWM's WebSDR.

This manual describes, section by section, all the controls visible on screen: their location, their function and how to use them. The more complex functions are accompanied by a thumbnail showing the window or panel as it actually appears in the interface.

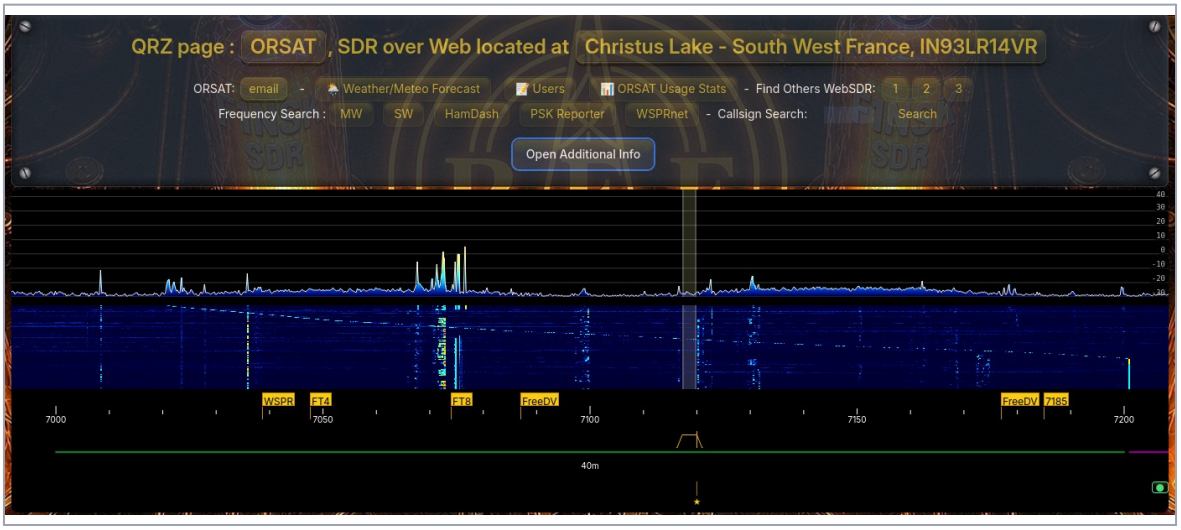
***Tip:** The interface is fully reactive: most settings (band, mode, filters, gain) apply instantly, with no need to confirm. Each user has their own tuning and their own settings, which are entirely private and have no effect on those of other users connected to the same receiver at the same time.*



Overview of the interface: top banner, spectrum/waterfall in the center, control columns on the left and right.

2. Top Banner

The banner at the very top of the screen identifies the station hosting the receiver and gathers all the related links and tools, arranged over three lines.



Top banner: station identification, related links, frequency and callsign search, and the button that opens the Additional Information menu.

2.1 Station identification line

The first line displays the site's name (or callsign) and its location, as defined by the administrator — for example *"QRZ page: ORSAT, SDR over Web located at Christus Lake - South West France, IN93LR14VR"*. The station name is a clickable link that opens the corresponding QRZ.com entry, useful for identifying the operator and the exact site of the receiver.

2.2 Second line: contact and site links

- **Station name: email** — opens an email message pre-addressed to the site administrator (questions, issue reports, etc.).
- **Weather/Meteo Forecast** — shows the local weather forecast for the receiver's site, useful for anticipating possible effects on propagation (storms, etc.).
- **Users** — opens a window listing the users currently connected to the receiver, possibly along with their listening frequency.
- **Station name Usage Stats** — opens a window with usage statistics for the receiver (load, number of connections, history, etc.).
- **Find Others WebSDR: 1, 2, 3…** — a numbered list of links to other WebSDR receivers recommended by the administrator (often other sites in the same community or station network).

2.3 Third line: frequency and callsign search

Frequency Search offers a series of external links to frequency databases and reception reports, to identify or cross-check what is being heard on the bands:

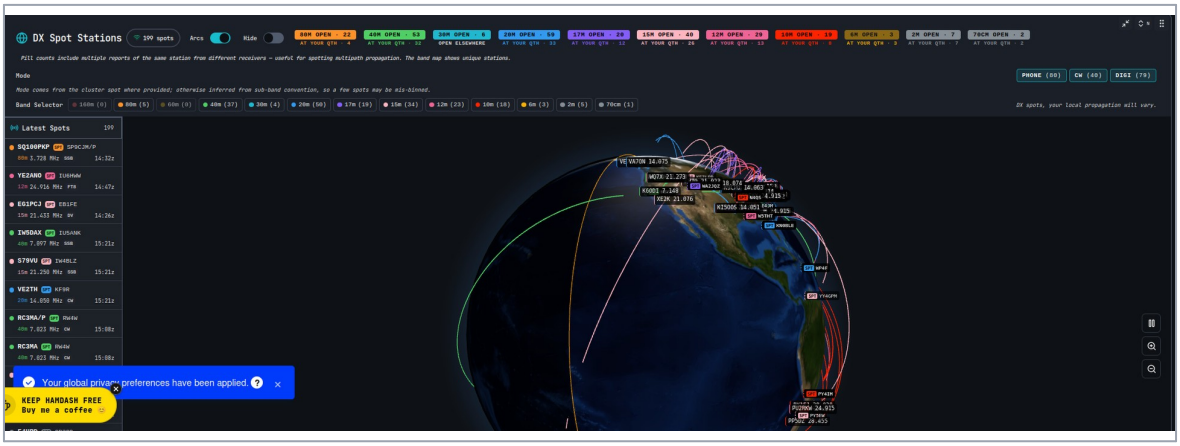
- **MW** — database of amplitude-modulated mediumwave stations.
- **SW** — database of shortwave stations (international broadcasting, utility stations).
- **HamDash** — community dashboard for amateur radio activity (hamdash.com), detailed below.
- **PSK Reporter** — community map of real-time reception reports for digital modes (FT8, PSK, etc.).

- **WSPRnet** — database of reception reports for the WSPR beacon mode.

Callsign Search offers an input field and a **Search** button for looking up a ham radio callsign directly (for example on QRZ.com or an equivalent database), without going through the station's own link.

HamDash in detail

HamDash (*hamdash.com*) is a community dashboard that aggregates in real time DX spots from the amateur radio cluster network and displays them on an interactive 3D globe, with arcs representing links between transmitting and receiving stations: an excellent way to gauge band conditions at a glance before starting to listen on ORSAT.



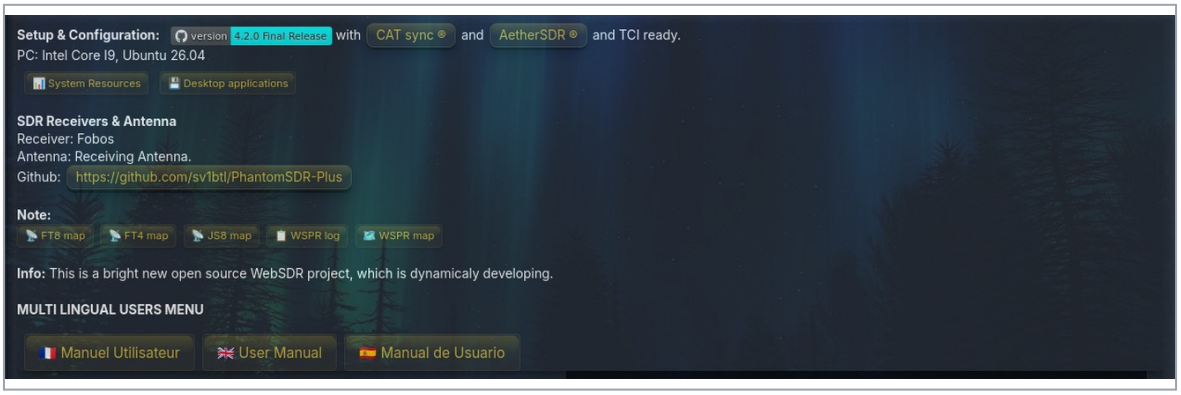
HamDash interface: 3D globe of DX spots, band-status tiles and a list of the latest spots.

- **Band-opening tiles** — at the top of the screen, one tile per band (80 m to 70 cm) shows at a glance whether the band is open, how many spots have been logged, and whether it is open “at your QTH” (from the location entered) or only elsewhere in the world.
- **Latest Spots** — left-hand column listing the most recent spots (callsign, spotter, band, frequency, mode, UTC time), clickable to center the globe on the corresponding link.
- **Globe and arcs** — each arc connects the spotted station to the station that spotted it; the **Arcs** toggle shows or hides them, and **Hide** temporarily hides the labels to declutter the display. The globe can be rotated and zoomed with the mouse.
- **Mode** — filters the displayed spots by mode family: **Phone** (voice), **CW** or **Digi** (digital modes such as FT8).
- **Band Selector** — a row of per-band checkboxes (with the spot count in parentheses) to display only the bands of interest.
- **Tile spot count** — the number shown on each band tile can count multiple reports for the same station received by different receivers, which is useful for spotting multi-path propagation; the globe map itself only shows unique stations.

***Tip:** HamDash is an independent external service, unrelated to PhantomSDR: the link in section 2.3 simply opens its site in a new browser tab.*

2.4 Opening the Additional Information menu

The **Open Additional Info and User Manual** button, centered below these three lines, expands the “Additional Information” menu described in section 13 (user manuals, system resources, desktop applications, FT8/FT4/JS8/WSPR links, etc.).



Top of the expanded menu: server configuration, host PC, receiver and antenna, community links and the multilingual manuals menu.

This expanded menu opens with a **Setup & Configuration** banner that summarizes the server status:

- **version** — the PhantomSDR-Plus version number used by the site (with a link to the project's GitHub repository).
- **CAT sync** and **AetherSDR** — badges indicating that CAT synchronization and compatibility with the AetherSDR software are active on this receiver.
- **TCI ready** — confirms that the TCI server described in section 4.5 (TCI - CAT) is operational.
- **PC** — hardware specifications of the server hosting the receiver (processor, operating system).

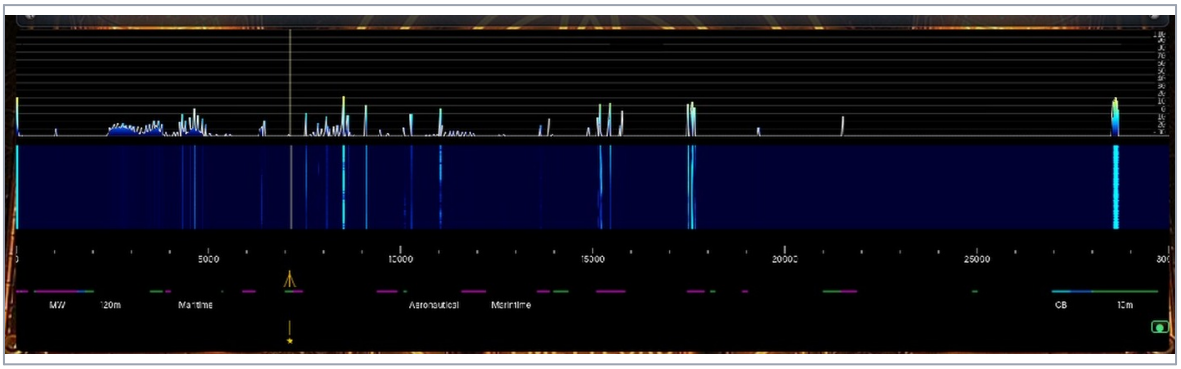
Next come the **System Resources** and **Desktop applications** buttons (detailed in sections 13.2 and 13.3), then an **SDR Receivers & Antenna** block specifying the SDR receiver model used, the receiving antenna, and a direct link to the **Github** repository of the PhantomSDR-Plus project. A **Note** block offers quick links to the community maps **FT8 map**, **FT4 map**, **JS8 map**, as well as a **WSPR log** and a **WSPR map** (see also section 13.5). An **Info** text briefly presents the PhantomSDR project as a modern, actively developed WebSDR, followed by the **Multi Lingual Users Menu** section, which groups the three user manual buttons (section 13.1).

> ***Tip:*** *It is precisely in this* ***Multi Lingual Users Menu*** *section that this user manual (and its French and Spanish versions) can be found: it is therefore accessible at any time from this expanded menu, without leaving the reception interface.*

3. Main screen: spectrum and cascade (waterfall)

The central area of the interface shows two complementary representations of the received signal:

- **The spectrum** (upper part) — displays the signal amplitude as a function of frequency, at the present instant, as a curve.
- **The cascade / waterfall** (lower part, scrolling) — displays the spectrum's history over time, each horizontal line representing one instant, with color conveying the intensity of the signal received at each frequency.



Spectrum and waterfall with the frequency scale and band/service markers below the waterfall.

Both displays are interactive: clicking or dragging positions the frequency cursor, the mouse wheel zooms in or out, and a horizontal drag scrolls the visible frequency window without changing the listening frequency.

- The vertical cursor indicates the currently tuned frequency.
- The displayed bandwidth (span) can be narrowed for a close look at a signal, or widened to monitor an entire portion of a band.
- The waterfall colors are adjustable (see section 10 — Waterfall Controls).

***Tip:** Double-clicking on the spectrum generally re-centers the display on the clicked frequency, handy for quickly exploring a band without losing the current tuning.*

4. Band, modes and reception filters

This family of controls, arranged around the frequency dial, lets you choose the amateur or broadcast band, the demodulation mode, the IF filter width, as well as switching between the two VFOs and linking with external control software (TCI/CAT).

4.0 Dial, S-meter and status indicators

The central panel displays the UTC and local time, the tuned frequency in digits (e.g. *007,120.00*), an analog S-meter graduated from S1 up to +60 dB above S9, and below it four instantaneous readings: **dBm** and **dBµV** (received signal level), **SNR** (signal-to-noise ratio in dB) and **NF** (noise floor, background noise level).



Central dial: time, frequency, S-meter and status indicators below the frequency.

Just below the frequency are the current band name, the active VFO, the mode and the filter width (e.g. *Current Band: 40m — VFO A | LSB | 2.70 kHz*), followed by a row of status indicators that light up in color when the corresponding function is active: **MUTED** (audio muted), **SQ** (squelch), **AI** (AI audio filter), **NR**, **NB**, **NS**, **AN** (noise filters, see section 5.3) and **CTCSS** (sub-tone decoding, useful in FM).

Tip: *This row of indicators is a great way to check at a glance which processing is currently active on the channel, without having to reopen each settings panel.*

Tip: *Clicking on the S-meter dial changes its color, letting you adapt it to your visual preferences or improve its readability depending on screen brightness.*

4.1 Band selector

The **Bands** button opens a window listing all the bands available on the receiver (for example 80 m, 40 m, 20 m, 17 m, etc., as well as international broadcast bands where applicable). Clicking on a band tunes the receiver directly to it, generally to a default frequency representative of the band (band center or the usual calling segment).



Band selection window.

> ***Tip:*** *Bands that are grayed out or missing from the list are not covered by the receiver's hardware or have been disabled by the site administrator.*

4.2 Demodulation modes

The **Modes** button opens the list of available demodulation modes: **USB**, **LSB**, **CW**, **AM**, **QUAM**, **FM**, plus two additional modes, **RADEL** and **RADEU**, on a second row. The chosen mode determines how the signal is demodulated, and therefore which IF filter is offered by default.

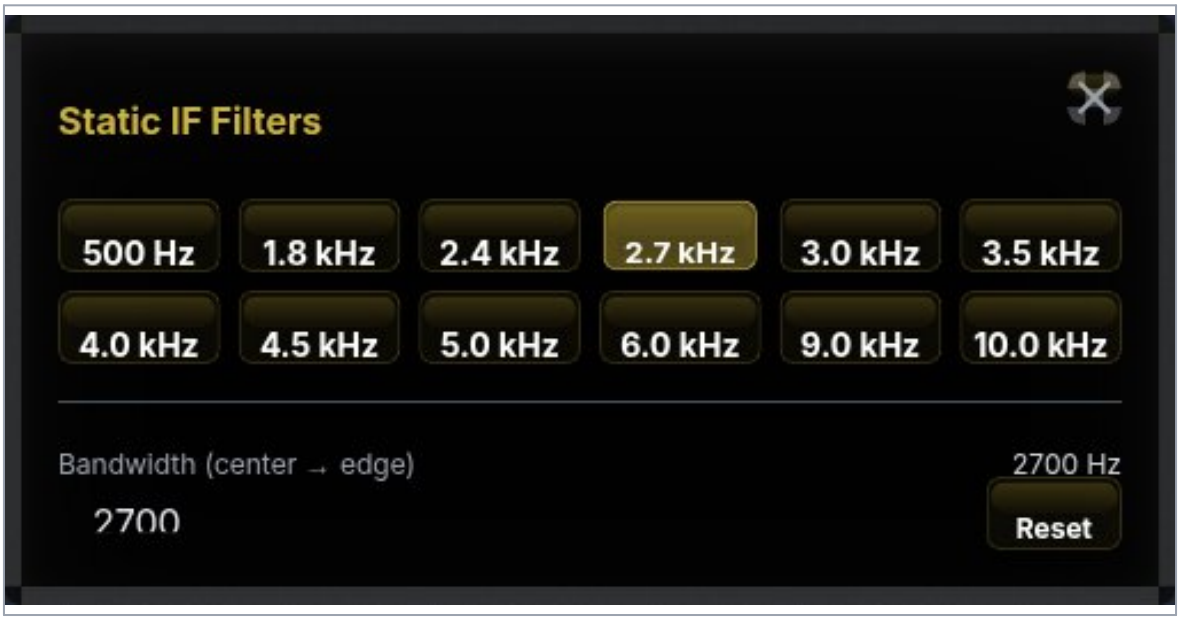


Demodulation mode selection window.

- **USB/LSB** — single sideband, used respectively above and below 10 MHz by convention on the amateur bands.
- **CW** — telegraphy, with a narrow filter suited to Morse reception.
- **AM** — conventional amplitude modulation.
- **QUAM** — a quadrature AM demodulation variant.
- **FM** — frequency modulation, used mainly on VHF/UHF.
- **RADEL / RADEU** — used to decode the **FreeDV** digital voice mode (lower/upper sideband), an open-source digital voice mode designed for shortwave use.

4.3 IF filters

The **IF Filters** button opens a window for adjusting the passband width of the filter applied to the demodulated signal, as well as, depending on the mode, the filter offset. A narrower filter reduces noise and adjacent interference but may cut off part of the wanted signal; a wider filter gives a more natural-sounding audio at the cost of reduced selectivity.



IF filter adjustment window.

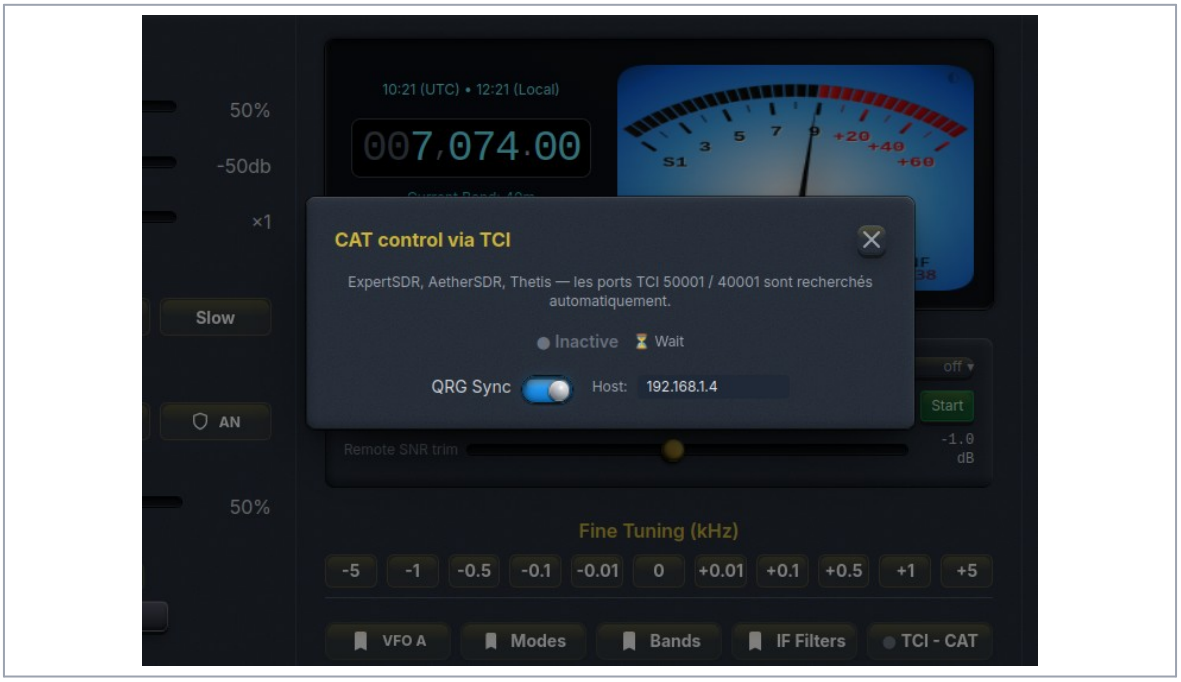
> ***Tip:*** *In CW, a very narrow filter (a few hundred Hz) markedly improves the signal-to-noise ratio on weak signals.*

4.4 VFO A / B

A VFO A/B toggle button lets you store a frequency on a second VFO and quickly switch between two listening frequencies, handy for monitoring a calling frequency while exploring the band on the other VFO.

4.5 TCI - CAT

The **TCI - CAT** button opens the “CAT control via TCI” window, which lets you synchronize the web interface with third-party software running on the user's computer (logging software, decoder, transceiver control software), via the TCI protocol. The **ExpertSDR**, **AetherSDR** and **Thetis** software packages are explicitly supported, with the interface automatically searching for their usual TCI ports (50001 and 40001).



“CAT control via TCI” window.

- **Status** — the window shows *Inactive* or *Wait* depending on whether a link is established or waiting for the third-party software to connect.
- **QRG Sync** — toggle that enables frequency (QRG) synchronization between the web interface and the remote software: a frequency change on either side is automatically reflected on the other.
- **Host** — field showing the IP address to use on the third-party software side to connect (for example *192.168.1.4*), useful when several network interfaces are available on the server machine.
- **Start button** — starts the local TCI server listening, ready to accept the connection from the third-party software.

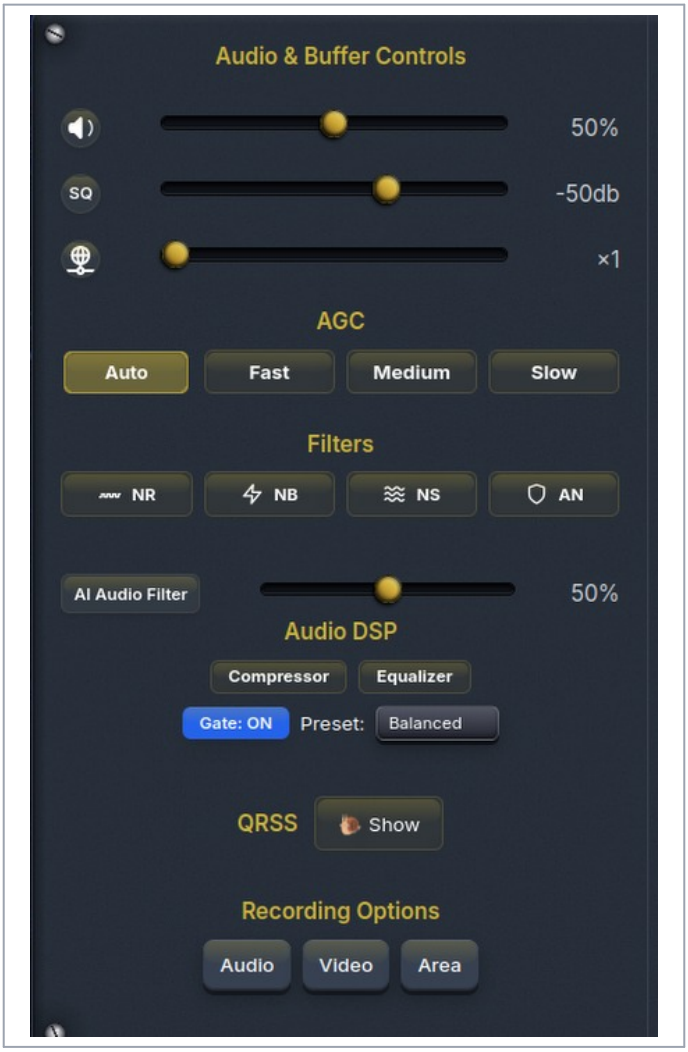
***Tip:** This function is used from software installed on the user's computer, on the same network as it, or reachable from it: it does not replace a direct CAT connection to an actual transceiver, but synchronizes the control software with the tuning done in the browser.*

4.6 Fine Tuning

The row of **Fine Tuning (kHz)** buttons lets you shift the frequency in fixed steps, in either direction, without having to retype the whole frequency: **-5, -1, -0.5, -0.1, -0.01, 0, +0.01, +0.1, +0.5, +1, +5** (in kHz). The central **0** button returns instantly to a zero step. The small steps (±0.01 and ±0.1 kHz) are particularly useful for precisely centering a CW or narrow digital signal in the filter, while the wider steps (±1 and ±5 kHz) let you jump quickly from one signal to another within the same band.

5. Left-hand audio control column

The left-hand column gathers all the settings related to audio playback and processing: audio level, automatic gain, noise-reduction filters and audio signal processing (DSP) tools.



Left-hand column: Audio & Buffer, AGC, Filters, Audio DSP, QRSS and Recording Options.

5.1 Audio & Buffer Controls

This section groups three settings, from top to bottom:

- **Volume** (speaker icon) — adjusts the local playback volume level; clicking the icon mutes/unmutes the sound (Muted, shown in the status indicators, section 4.0).
- **SQ** (Squelch) — sets the squelch threshold, in dB: below this threshold, the audio is automatically muted, which masks background noise when no signal is present.
- **Network buffer size (globe icon)** — multiplier for the audio buffer (×1 by default). A larger value smooths out any dropouts caused by an unstable network connection, at the cost of higher audio latency; a smaller value reduces latency but makes micro-dropouts more likely over a poor-quality connection.

5.2 AGC (Automatic Gain Control)

The automatic gain control continuously adjusts the amplification of the received signal to maintain a consistent listening level, compensating for fast variations in signal strength. Four modes are offered as buttons: **Auto**, **Fast**, **Medium** and **Slow**.

- **Auto** — lets the receiver choose the response speed best suited to the received signal; this is the recommended default setting.
- **Fast** — reacts quickly to impulsive or highly variable signals (useful in CW) but can “pump” on some fluctuating signals.
- **Medium** — an intermediate compromise, suited to most SSB use.
- **Slow** — slow response, generally preferred for AM/broadcast for a more natural, stable rendering of voice or music.

5.3 Filters (NR / NB / NS / AN / AI Audio Filter)

This section brings together several noise- and interference-reduction filters, which can be applied individually or in combination:

- **NR** (Noise Reduction) — broadband, spectral-type noise reduction, attenuates continuous background noise without targeting any specific type of interference.
- **NB** (Noise Blanker) — suppresses brief impulsive noise (ignition noise, crackles), by “blanking” the detected pulses.
- **NS** (Notch/Squelch depending on context) — a notch or automatic squelch filter depending on the implementation, letting you cut an annoying carrier or mask the background noise when no signal is present.
- **AN** (Auto Notch) — automatically detects and attenuates fixed carriers (heterodynes) present in the audio channel.
- **AI Audio Filter** — a denoising filter based on an artificial-intelligence model, offering deeper noise reduction on voice than the classic NR.

***Tip:** These filters can be combined, but excessive use (several aggressive filters active at once) can degrade the intelligibility of the wanted signal: it is recommended to enable them one at a time, listening to the effect as you go.*

5.4 Audio DSP (Compressor / Equalizer / Gate / Presets)

The audio digital signal processing (DSP) module lets you fine-tune the final sound. The **Compressor** and **Equalizer** buttons each open a dedicated settings window.

- **Compressor** — reduces the dynamic range of the audio signal (the gap between weak and strong sounds), making listening more comfortable on variable signals.
- **Equalizer** — adjusts the balance between low/mid/high frequencies of the restored sound, for example to emphasize voice intelligibility or listening comfort in CW.
- **Gate** (Noise Gate) — mutes the audio when the level drops below a set threshold, useful for cutting background noise between words or transmissions.
- **Presets** — preconfigured DSP profiles (for example geared toward voice, CW or digital) letting you apply a recommended combination of settings with a single click.

Compressor window

An **Enable Compressor** toggle activates the module, with the following settings and a **Reset** button to return to the default values:

- **Threshold** — the level (in dB) above which compression starts to act.
- **Ratio** — the compression ratio applied beyond the threshold (for example 4:1).
- **Attack** — the compressor's response time when the signal exceeds the threshold (in ms).

- **Release** — the release time once the signal drops back below the threshold (in ms).
- **Knee** — smoothness of the transition around the threshold (in dB); a higher value softens the onset of compression.
- **Makeup** — the makeup gain applied after compression, to restore a comfortable listening level; the **Auto makeup** checkbox calculates this gain automatically from the chosen threshold and ratio.



Audio Compressor settings window.

Equalizer window

An **Enable Equalizer** toggle activates the module, with six frequency bands individually adjustable in gain (dB): **100 Hz, 350 Hz, 1 kHz, 2 kHz, 3 kHz** and **5 kHz**, and a **Reset** button to return to a flat response (0 dB across the board).

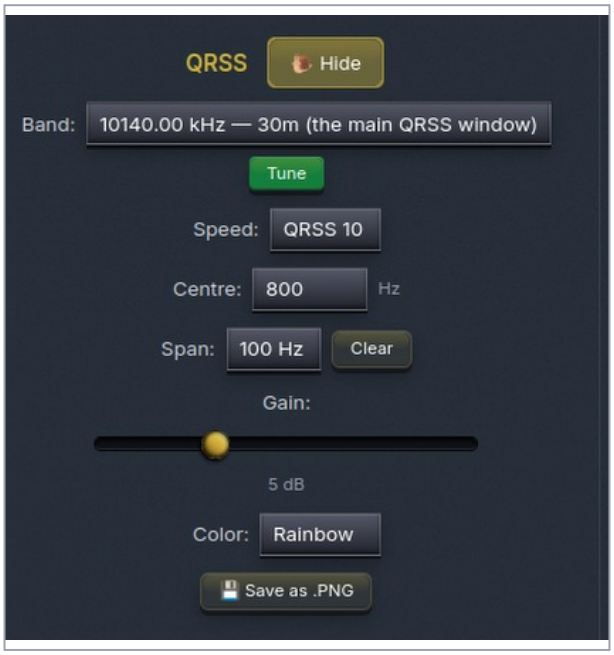


Six-band audio Equalizer settings window.

> ***Tip:*** *To bring voice out of the noise, try slightly attenuating the low frequencies (100–350 Hz) and slightly boosting the 1–2 kHz range, where speech intelligibility is concentrated.*

6. QRSS

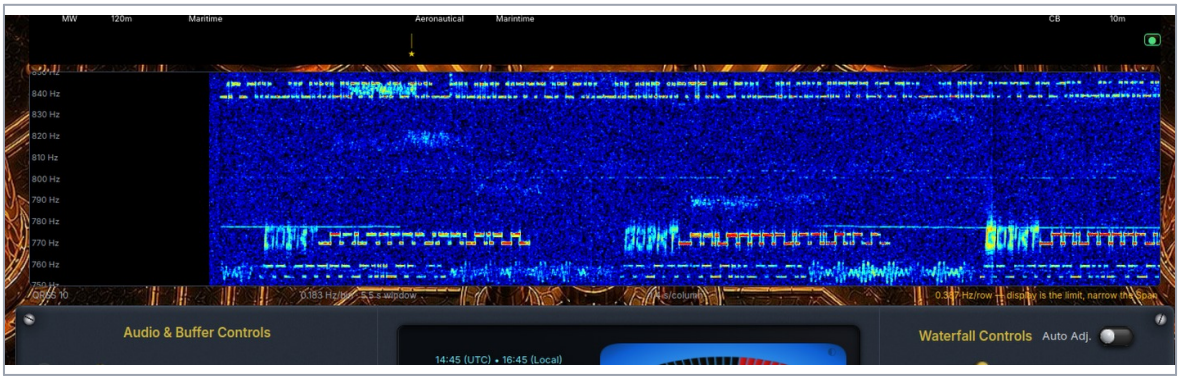
The **Show** button, next to the QRSS label, opens a dedicated settings panel and displays a visual decoder overlaid on the waterfall, designed for observing the very slow, very low-power signals typical of QRSS traffic (extremely slow Morse, several seconds per dot). The button becomes **Hide** once the overlay is displayed, to hide it again.



QRSS settings panel.

The QRSS panel offers:

- **Band** — selects a predefined QRSS beacon or band segment (for example *10140.00 kHz — 30m, the main QRSS window*), then the **Tune** button tunes the receiver directly to that frequency.
- **Speed** — the speed of the QRSS traffic being observed (for example *QRSS 10*, one Morse dot roughly every 10 seconds), which determines the resolution of the time-frequency analysis applied to the overlay.
- **Centre** — the central audio frequency of the observation window, in Hz (for example 800 Hz).
- **Span** — the width of the frequency window displayed around the center, in Hz (for example 100 Hz); the **Clear** button resets the accumulated display.
- **Gain** — a contrast/sensitivity slider for the overlay (in dB), to adjust the visibility of weak traces without drowning them in noise.
- **Color** — the overlay's color palette (for example *Rainbow*), independent of that of the main waterfall (section 10).
- **Save as .PNG** — saves an image of the displayed QRSS overlay, useful for archiving or sharing a particular reception.



QRSS overlay on the waterfall: audio frequency scale (Hz) on the left, decoded QRSS traces, and resolution indicators at the bottom (speed, Hz/bin resolution, window duration, Hz/line resolution).

Below the overlay, a row of technical indicators shows the current analysis resolution: the selected QRSS speed, the frequency resolution (Hz/bin) and the duration of the analysis window (in seconds), the time represented per column, and the vertical resolution (Hz/line) — the latter sometimes flagged as the display limit, in which case it is advisable to narrow the **Span** to gain definition.

***Tip:** Enable QRSS when monitoring a beacon or a station known to transmit in QRSS (usually in dedicated band segments, for example around 10,140 kHz on 30 m); the overlay stays active until it is hidden via the Hide button.*

7. Recording Options

This section offers three local recording buttons, corresponding to three different kinds of capture:

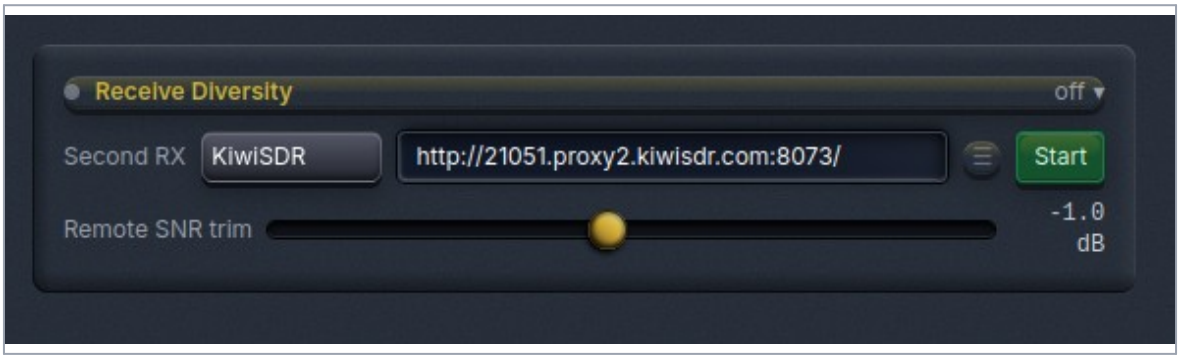
- **Audio** — records only the demodulated audio stream of the channel currently being listened to.
- **Video** — records a video capture (typically the waterfall and/or spectrum display over time), useful for keeping a visual record of an observed event.
- **Area** — delimits and records a specific area of the screen or of the spectrum/waterfall, to target a precise portion of the band rather than the whole display.

Once recording is stopped, a download link or button lets you retrieve the resulting file onto the user's computer.

Tip: *Remember to check the tuned mode and frequency before starting a recording: the recording captures exactly what is heard, including filter and DSP settings.*

8. Receive Diversity

Unlike local antenna diversity, the **Receive Diversity** function of PhantomSDR-Plus combines the local receiver with a **second, remote receiver accessible over the network** (typically a public or private KiwiSDR), in order to improve the signal-to-noise ratio or reduce the effect of selective fading by combining two geographically different listening points.



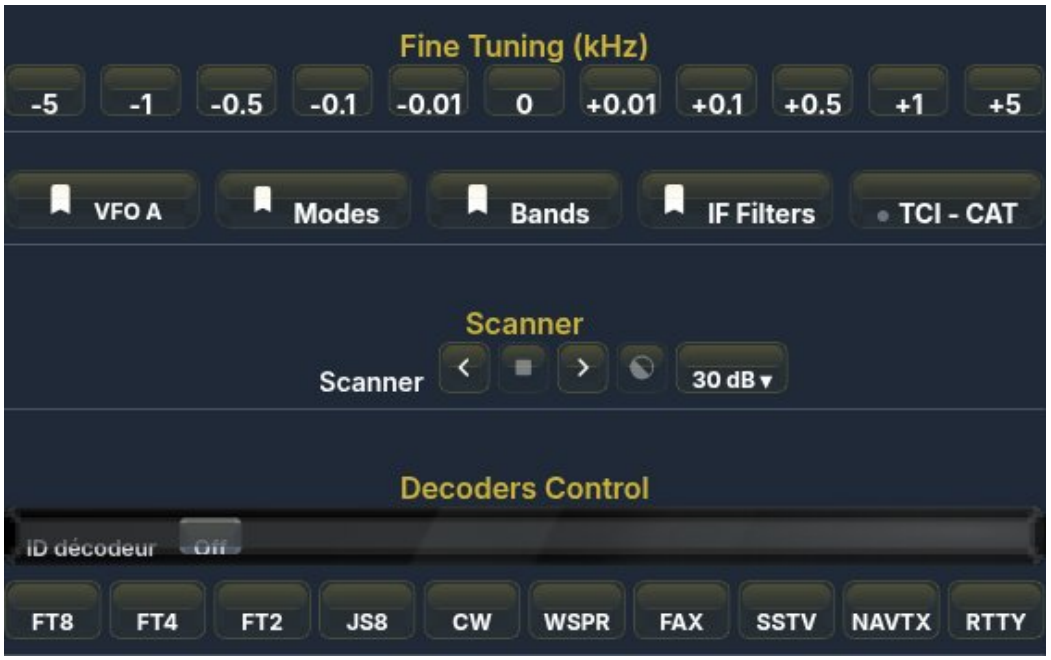
Expanded Receive Diversity panel: second receiver (Second RX), address and Start button.

- **On/off toggle** — enables or disables receive diversity (shown *off* by default, at the top of the panel).
- **Second RX** — the type of remote receiver to combine with (for example *KiwiSDR*).
- **Address field** — the URL of the remote receiver to use as a second source, for example *http://21051.proxy2.kiwisdr.com:8073/*.
- **Start button** — connects to the designated remote receiver and starts combining the two streams.
- **Remote SNR trim** — a slider that compensates for a level difference (in dB) between the local and remote receivers, so as to balance their respective contribution to the combined signal.

***Tip:** The chosen remote receiver must cover the same band and be publicly available (or reachable from the server); performance depends heavily on the quality of its own reception and the network latency to it.*

9. Scanner and Decoders

This row of controls groups two distinct functions: automatic frequency scanning (Scanner) and control of the built-in digital decoders.



Fine Tuning / Scanner / Decoders row.

9.1 Scanner

The Scanner automatically sweeps the bands, using three buttons (previous, stop, next) and a circular scan-mode button, stopping automatically as soon as a signal above the set threshold is detected.

- **■ / ■ / ■ buttons** — step back one, stop, or step forward one in the scan; a long press starts continuous scrolling.
- **Circular button** — toggles the scan mode (for example continuous scanning or cycling through the bookmark list, see section 12).
- **dB threshold (e.g. “30 dB”)** — a dropdown menu setting the signal level, above the background noise, above which the scanner considers a frequency active and stops on it.

9.2 Decoders Control

The digital decoders panel offers ten modes that can be enabled with a single click, as a row of buttons: **FT8, FT4, FT2, JS8, CW, WSPR, FAX, SSTV, NAVTX** and **RTTY**. A master toggle (“Decoder ID”, On/Off) enables or disables the whole decoding module.



Decoders panel: mode selection, audio waveform with decode markers, and a table of decoded messages.

Once a mode is selected (for example *CW* or *FT8*), the panel shows:

- **An audio waveform** with yellow vertical markers flagging each successful decode ("yellow marks = decoded spots"), letting you visually spot where activity lies within the audio passband.
- **Farthest Distance** — the distance to the most distant decoded contact since the start of the session (in km), a quick indicator of propagation conditions.
- **Sync offset** — a slider to correct the synchronization offset, with an **Auto** checkbox letting the decoder adjust this offset automatically.
- **Table of decoded messages** — one row per decode, with the columns **MODE** (mode used), **DB** (decoded signal level), **HZ** (the audio frequency of the signal within the passband), **DT** (time-sync offset), **MESSAGE** (decoded content: callsigns and report), **LOCATOR** (the station's Maidenhead grid locator, if transmitted) and **DIST** (estimated distance to that station).

Tip: *To automatically forward these decodes to logging software or a community map installed on your own computer, see section 4.5 (TCI - CAT), which links the web interface to third-party software.*

10. Waterfall Controls

Located in the right-hand column, these controls adjust the appearance and sensitivity of the spectrum and waterfall display, independently of the audio signal itself: they only change the visual representation.

- **Auto Adj.** (toggle) — automatically calibrates the waterfall's Min/Max levels based on the ambient background noise, handy for getting a readable display with a single click, with no manual adjustment.
- **Min** and **Max** — sliders that manually set, in dB, the low and high bounds of the intensity being represented; narrowing the gap between Min and Max increases the visual contrast of weak signals, at the risk of clipping strong ones.
- **Zoom** (×1 slider and *In* / *Out* / *Max* / *Min* buttons) — horizontal zoom of the waterfall and spectrum over the displayed frequency range; the **To Band** button automatically re-frames the display over the full width of the current band.
- **Waterfall Display Color** — dropdown menu for selecting the color palette used to represent signal intensity on the waterfall (the name of the active palette, for example *PhantomSDR*, is shown in the menu).

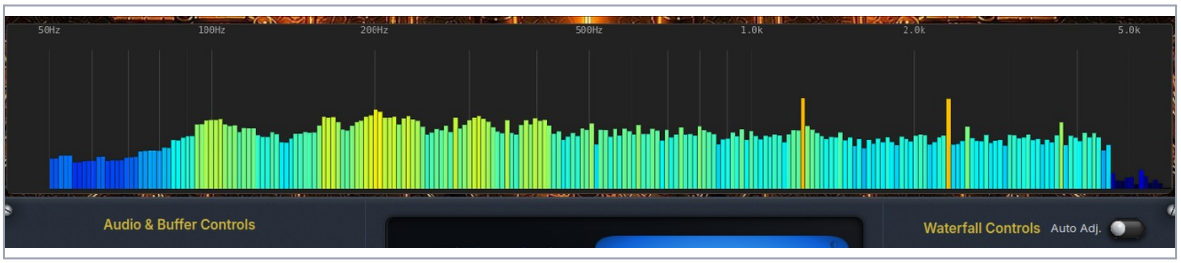


Waterfall Controls panel: Auto Adj., Min/Max, Zoom and Waterfall Display Color.

> ***Tip:*** *A good Min/Max setting (or using Auto Adj.) is often the difference between an unreadable waterfall, either all black or all white, and a display where weak signals stand out clearly from the background noise.*

11. Audio spectrogram

The **Audio Spectrogram — Show** button, located below the waterfall controls, shows or hides an audio spectrum analyzer separate from the main RF waterfall: a bar-chart histogram representing, in real time, the amplitude of the demodulated audio signal on a logarithmic frequency scale (from roughly 50 Hz to 5.0 kHz). It is particularly useful for visualizing the structure of digital signals (FT8, RTTY, PSK, etc.) directly within the audio band, or for visually spotting a CW signal or a precise tone within the current filter passband.

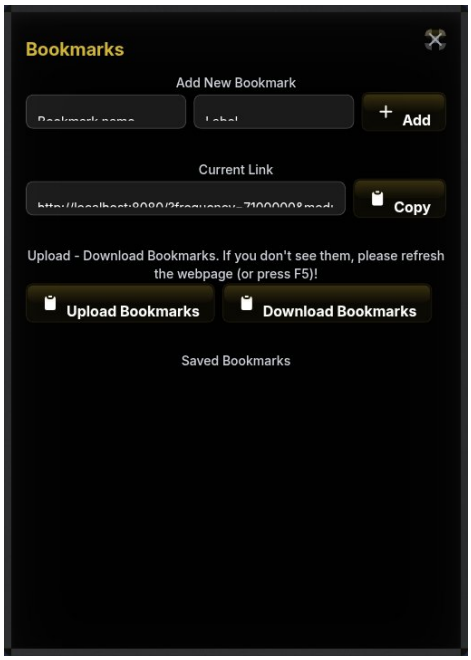


Audio spectrum analyzer: amplitude by frequency, logarithmic scale from 50 Hz to 5.0 kHz.

- Helps precisely position the IF filter on a digital signal visible in the audio.
- Complements the main RF waterfall, which covers a much wider frequency range but with less detail within the audio band itself.
- The **Show** button changes to **Hide** once the spectrogram is displayed, to close it again and free up screen space.

12. Bookmarks

The Bookmarks window lets you save frequencies of interest so you can return to them quickly with a click, and save that list as a reusable file.

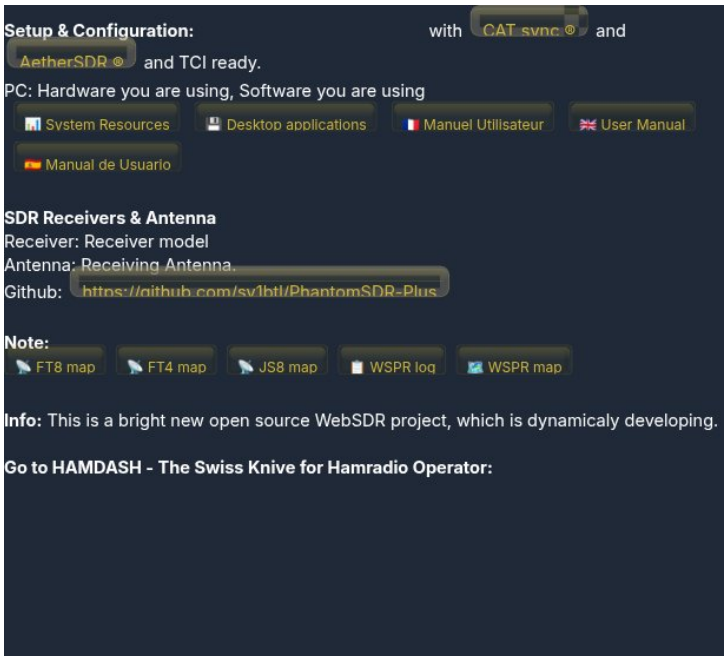


Bookmarks window.

- **Add New Bookmark** — adds a new bookmark from the *Bookmark name* (the name given to the bookmark) and *Label* (a short tag shown on the waterfall) fields, confirmed with the **+ Add** button; the bookmark stores the current frequency and tuning parameters.
- **Current Link** — shows a direct link (URL) pointing to the interface's current frequency and settings, copyable with one click via the **Copy** button; handy for sharing a precise reception with someone else, independently of the bookmark list itself.
- **Upload Bookmarks / Download Bookmarks** — exports the current bookmark list to a downloadable file, or imports a previously saved bookmark list; after an import, the interface notes that you may need to refresh the page (F5) to see the new bookmarks appear.
- **Saved Bookmarks** — a list of already-saved bookmarks; clicking an entry tunes the receiver to the corresponding settings.

13. “Additional Information” menu

This menu, generally collapsible and accessible from a dedicated button in the interface, gathers a set of related tools that don't directly concern the reception itself: documentation, system resources, useful external links and information about the station.



“Additional Information” menu, including the user manual buttons.

13.1 User manuals

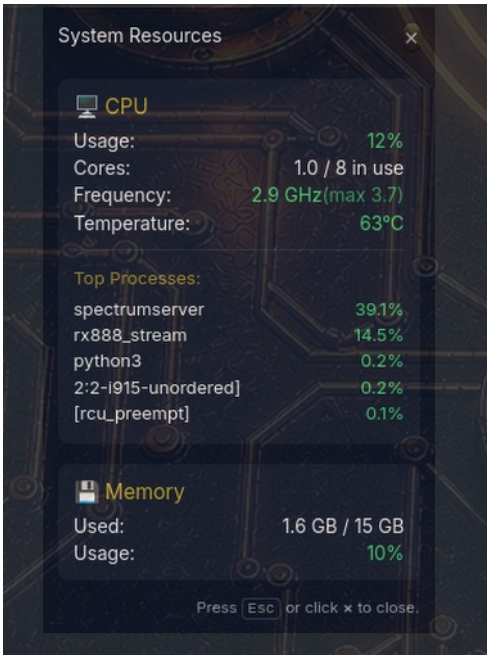
Three buttons give access to this user manual, each in a different language, identifiable by their flag:

- **[FR] Manuel Utilisateur** — opens this document in French.
- **[EN] User Manual** — opens the English version of the manual.
- **[ES] Manual de Usuario** — opens the Spanish version of the manual.

Clicking any of these buttons opens a dedicated window displaying the corresponding PDF document, with a native vertical scrollbar letting you browse it in full. The window closes via the “×” button or the Escape key.

13.2 System resources

This button opens a window showing the load status of the server hosting the receiver (CPU usage, memory, possibly network bandwidth or number of active connections), useful for understanding any interface slowdowns.



Server system resources window.

13.3 Desktop applications

A “Desktop applications” link points to third-party client software that lets you connect to the receiver from a native application installed on the computer (rather than from the browser), for users who prefer that mode of use.

13.4 SDR receiver information

This entry displays the technical specifications of the SDR receiver used by the site (hardware model, frequency range covered, antenna(s) used, approximate location, etc.), useful for gauging expected performance depending on the band being listened to.

13.5 FT8 / FT4 / JS8 / WSPR links

Quick links point to online decoding maps (such as PSK Reporter or equivalent) for the digital modes FT8, FT4, JS8 and WSPR, letting you see in real time the station's reception reports on these modes, reported by the worldwide community.

14. DX Cluster

The DX Cluster panel shows in real time the spots posted by the worldwide ham radio community on active DX stations, generally indicating the spotted callsign, frequency, mode, the spotter and any comment.

- Clicking a spot generally tunes the receiver directly to the corresponding frequency, making it a quick way to “chase DX” flagged by other operators.
- The feed can generally be filtered (by band, mode or continent) depending on the options offered by the interface.

15. Tips, troubleshooting and best practices

15.1 Listening quality

- Start with an IF filter suited to the mode (wide for AM/FM, narrow for CW) before enabling the noise-reduction filters (NR/NB/NS/AN).
- Enable the noise filters gradually, one at a time: combining too many can degrade intelligibility more than it improves it.
- Adjust the waterfall's Min/Max (or use the automatic adjustment) to clearly tell weak signals apart from the background noise.

15.2 In case of audio dropouts or latency

- Increase the audio buffer size (section 5.1) to smooth out an unstable network connection, at the cost of a slight extra delay.
- Check System Resources (section 13.2): a high server load can explain slowdowns on the interface side.

15.3 Collaborative use

- Several users can connect simultaneously to the same receiver: each has their own tuning (frequency, band, mode, filter) and their own audio settings (volume, DSP, noise filters), which are entirely private and have no effect on those of other users connected at the same time.
- The DX Cluster (section 14) makes it easier for users connected at the same time to coordinate with one another.

End of user manual — English version. For any further questions, please contact the administrator of the site hosting this receiver.